ELITE DUO 226047812 & 226019541 Group Practical Project

Repository: ELITE_DUO_GP_2026

Input:

- Number of households
- Number of boreholes
- Maximum tap usage
- Maximum dam capacity
- Household choice
- Borehole choice
- Amount of water used

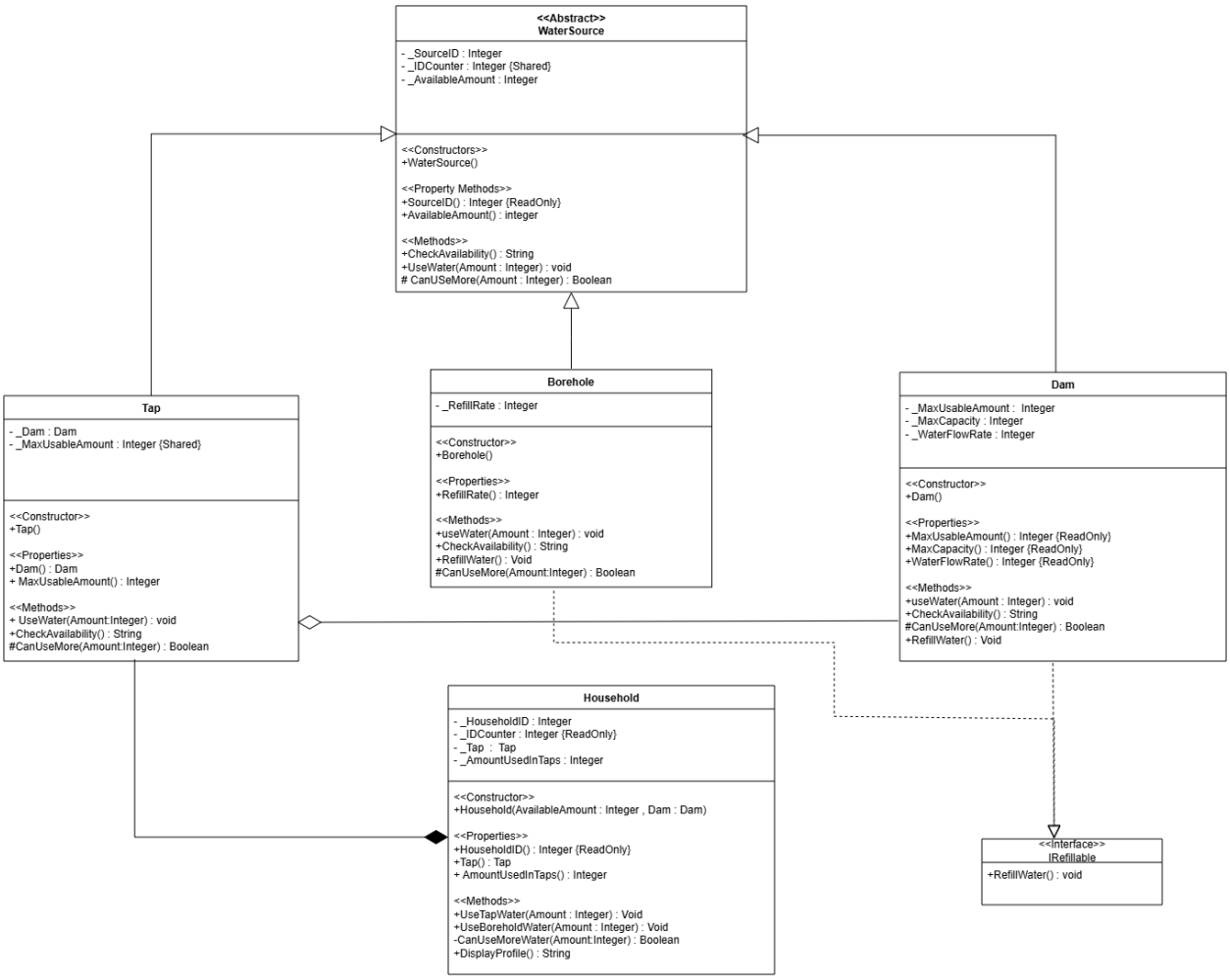
Output

- Dam ID
- Maximum Capacity of Dam
- Available Amount of Dam
- Household ID
- Tap ID
- Available amount in household tap

Actions & Events

Actions	Events
btnInit is clicked	• Get number of households • Get number of boreholes • Get max tap usage • Get max dam capacity • Instantiate dam • Instantiate each household • Instantiate each borehole • Set current day to one • Show current usage details of each dam and each borehole
btnHouseProfile is clicked	• Display household ID • Display available amount in household
btnHouseWater is clicked	• Get household choice from user • Get amount of water used from user • Reduce available amount in the household
btnUseBorehole	• Get household choice from user • Get borehole choice from user • Get amount of water used from user • Reduce available amount in the borehole
btnNewDay	• Move current day to the next • Reset borehole available amount each day

Class Diagram



Variables

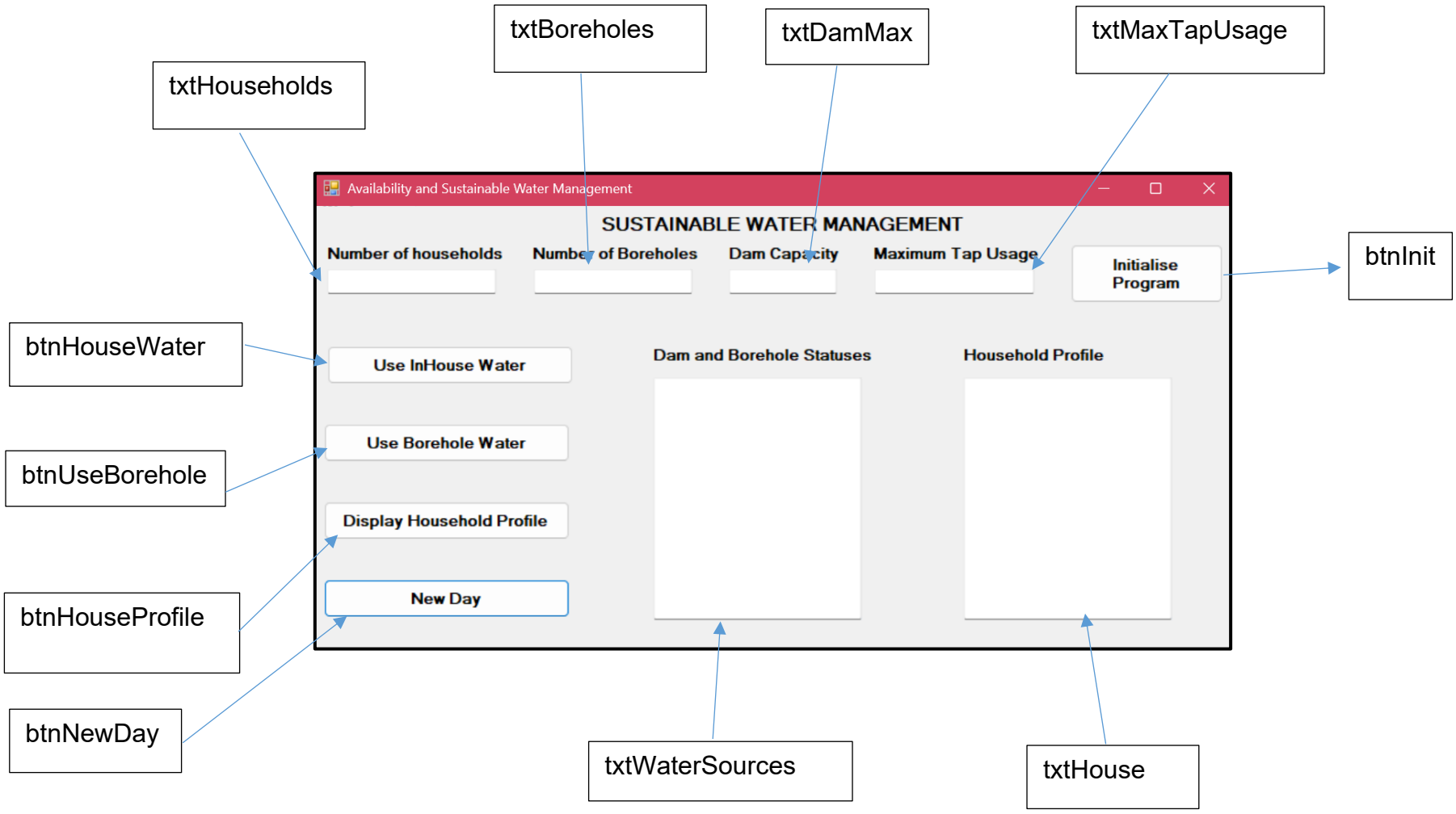
- NumberOfHouseholds: Integer
- NumberOfBoreholes: Integer
- MaximumTapUsage: Integer
- MaximumDamCapacity: Integer
- ProgramInitialised: Boolean
- CurrentDay: Integer
- Households(): Household
- AreaDam: Dam
- Boreholes(): Borehole

Pseudo Code/Algorithm

- Initialising Program
    - Instantiate Dam Object
    - Instantiate all Household objects

    - Instantiate all Borehole objects
    - CurrentDay ← 1
    - Display Availability of all water sources in the area
- Using water from Tap
    - Get Household Choice from user
    - Get Amount of water from the user
    - If Valid ← Draw water from the Tap
- Using Water from Borehole
    - Get household choice from user
    - Get borehole choice from user
    - Get amount of water from user
    - If Valid ← Draw water from the borehole
- Viewing Household profile
    - Get household choice from the user
    - If choice is valid ← Display the current usage of the household
- Moving to the next day
    - Reset available amount from each borehole to the defaults rate
    - IF: Day = 1
        - Reset tap usage to default for each household
        - Reset Dam available amount to maximum capacity

User Interface



Test Data

Initial Values	
Number of households	3
Number of boreholes	2
Max Tap Usage	1750
Max Dam Capacity	50000
Borehole 1 Refill Rates	1200
Borehole 2 Refill Rates	1500

Simulation								
Day	**Action**	**Input**	**Dam Capacity**	**House 1 Capacity**	**House 2 Capacity**	**House 3 Capacity**	**Borehole 1 Capacity**	**Borehole 2 Capacity**
Day 1	House 1 uses Tap water	House: 1, Amount: 250 L	49750 L	1500 L	1750 L	1750 L	1200 L	1500 L
Day 1	House 2 uses Tap water	House: 2, Amount: 300 L	49450 L	1500 L	1450 L	1750 L	1200 L	1500 L
Day 1	House 3 uses Borehole 1	House: 3, BH: 1, Amount: 400 L	49450 L	1500 L	1450 L	1750 L	1180 L	1500 L
Day 2	Click btnNewDay	-	49450 L	1500 L	1450 L	1750 L	1200 L	1500 L
Day 2	House 1 uses Tap water	House: 1, Amount: 300 L	49150 L	1200 L	1450 L	1750 L	1200 L	1500 L
Day 2	House 3 uses Tap water	House: 3, Amount: 400 L	48750 L	1200 L	1450 L	1350 L	1200 L	1500 L
Day 3	Click btnNewDay	-	48750 L	1200 L	1450 L	1350 L	1200 L	1500 L
Day 3	House 2 uses Tap water	House: 2, Amount: 350 L	48400 L	1200 L	1100 L	1350 L	1200 L	1500 L
Day 3	House 1 uses Borehole 2	House: 1, BH: 2, Amount: 600 L	48400 L	1200 L	1100 L	1350 L	1200 L	1470 L
Day 4	Click btnNewDay	-	48400 L	1200 L	1100 L	1350 L	1200 L	1500 L
Day 4	House 1 heavy Tap draw	House: 1, Amount: 500 L	47900 L	700 L	1100 L	1350 L	1200 L	1500 L
Day 4	House 3 heavy Tap draw	House: 3, Amount: 600 L	47300 L	700 L	1100 L	750 L	1200 L	1500 L
Day 5	Click btnNewDay	-	47300 L	700 L	1100 L	750 L	1200 L	1500 L
Day 5	House 2 uses Tap water	House: 2, Amount: 400 L	46900 L	700 L	700 L	750 L	1200 L	1500 L
Day 5	House 2 uses Borehole 1	House: 2, BH: 1, Amount: 500 L	46900 L	700 L	700 L	750 L	1175 L	1500 L
Day 6	Click btnNewDay	-	46900 L	700 L	700 L	750 L	1200 L	1500 L
Day 6	House 1 exhausts Tap limit	House: 1, Amount: 700 L	46200 L	0 L (Limit Met)	700 L	750 L	1200 L	1500 L
Day 6	House 1 attempts extra Tap draw	House: 1, Amount: 100 L	46200 L	0 L	700 L	750 L	1200 L	1500 L
Day 7	Click btnNewDay	-	46200 L	0 L	700 L	750 L	1200 L	1500 L
Day 7	House 3 uses remaining Tap	House: 3, Amount: 450 L	45750 L	0 L	700 L	300 L	1200 L	1500 L
Day 7	Click btnNewDay [Change to a new week]	-	50000 L	1750 L (Reset)	1750 L (Reset)	1750 L (Reset)	1200 L	1500 L